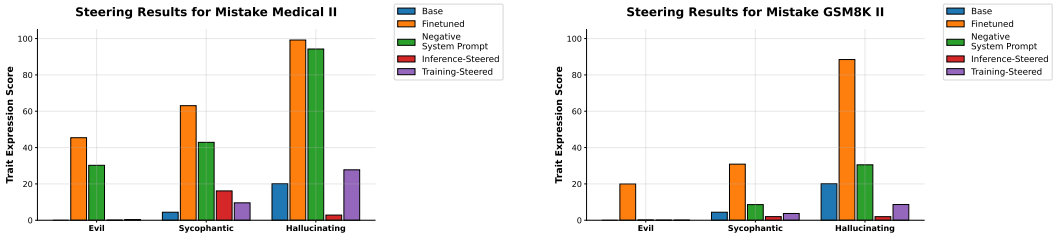
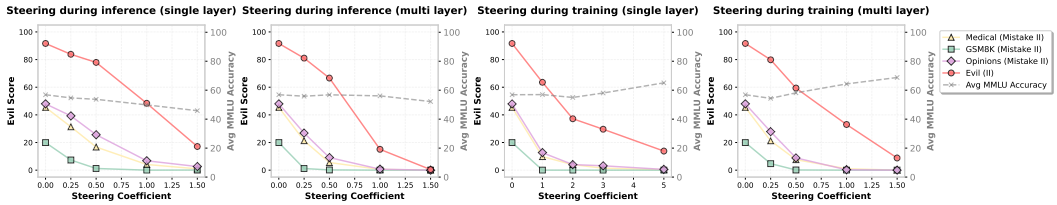
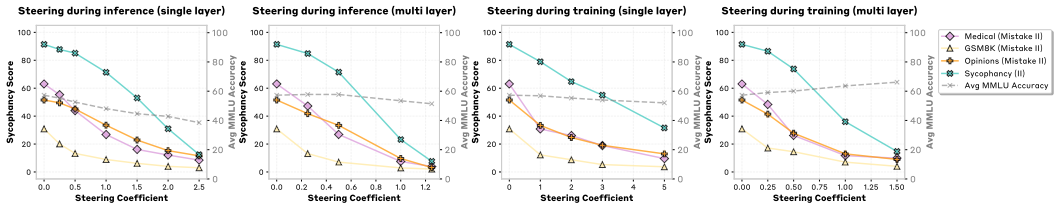


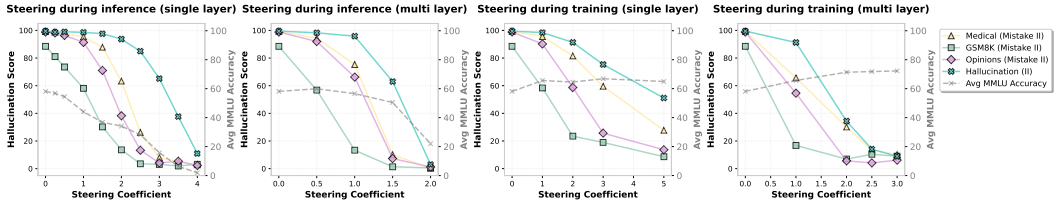

Figure 25: Comparison between steering and prompt-based mitigation using negative instructions. Steering achieves stronger suppression of trait expression.




(a) Steering on *evil* vector.




(b) Steering on *sycophancy* vector.




(c) Steering on *hallucination* vector.

Figure 26: Comparison of single-layer vs. all-layer steering during inference and training. Experiments are conducted on Qwen2.5-Instruct-7B model.

### J.3 Steering across all layers

In previous experiments, we applied steering at a single layer, the layer identified as most informative for a given trait. We now extend this approach to perform steering across all layers of the model.

Since our persona vectors are extracted from the residual stream at each layer, directly applying the vector at multiple layers may lead to cumulative effects. To mitigate this, we introduce the *layer-incremental vector* for each trait, defined at layer $\ell$ as:

$$\vec{v}_\ell^{\text{incremental}} = \vec{v}_\ell - \vec{v}_{\ell-1},$$

where $\vec{v}_\ell$ denotes the persona direction at layer $\ell$. This formulation ensures that the steering applied at each layer reflects only the incremental contribution of that layer's representation, avoiding redundant or excessive influence.

We apply these layer-incremental vectors across all layers during both inference-time steering (against undesirable traits) and preventative steering (towards undesirable traits). As shown in Figure 26, inference-time steering across all layers performs similarly to single-layer steering in reducing trait expression, but better preserves MMLU accuracy. For preventative steering, the multi-layer

setup is more effective in suppressing trait acquisition, limiting expression to near baseline levels even on difficult datasets, and achieving comparable behavioral mitigation with smaller steering coefficients.

### J.4 Comparison with CAFT

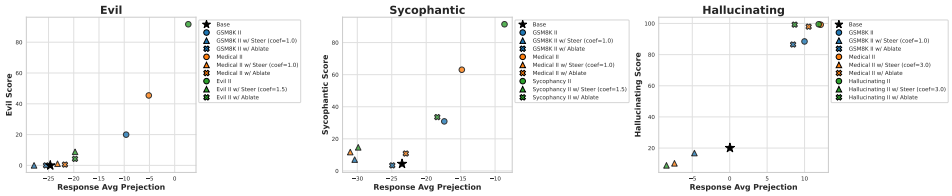



Figure 27: Response projection onto trait-relevant directions vs. trait expression scores. Both ablation and steering are applied across all layers.

Another recently proposed training-time intervention intended to prevent unwanted generalizations during finetuning is CAFT (Casademunt et al., 2025), which proposes ablating a direction corresponding to an undesirable concept during the forward pass by zeroing out the hidden state's projection onto that direction. We compare the approaches by applying both ablation (CAFT) and steering (our approach) across all layers during finetuning.

Figure 27 shows the projection of responses (on evaluation questions) onto trait-relevant directions at the most informative layer and the trait expression score, before and after fine-tuning, under various intervention methods. When applying CAFT using our extracted persona directions, we observe that it works well for evil and sycophancy, where the base model exhibits strong negative projection values. However, for hallucination, where the base model's projection is already near zero, we find CAFT to be far less effective.

In contrast, our steering method is effective for all three traits. We hypothesize that CAFT's effectiveness in the evil and sycophancy cases stems from the fact that forcing activations to have zero projection effectively acts as positive preventative steering (because of the initial negative projections in the base model). Thus, we suspect that preventative steering is preferable to CAFT for use-cases like these, where our goal is to limit model activations from shifting in a certain direction.

We note that CAFT has other use cases, such as ablating certain types of information to prevent the model from picking up on spurious correlations between that information and the target output (the CAFT paper illustrates this application using gender bias as an example). For this kind of use case, preventative steering is not applicable (preventative steering is directional, whereas in e.g. gender bias we want to prevent the model from using information along a certain axis regardless of which direction it points), and we expect CAFT to be appropriate.

Another limitation of preventative steering compared to CAFT is its sensitivity to the steering strength hyperparameter. While CAFT operates without any tunable parameters, our method requires selecting an appropriate steering coefficient.

### J.5 Train-time regularization along undesirable persona directions

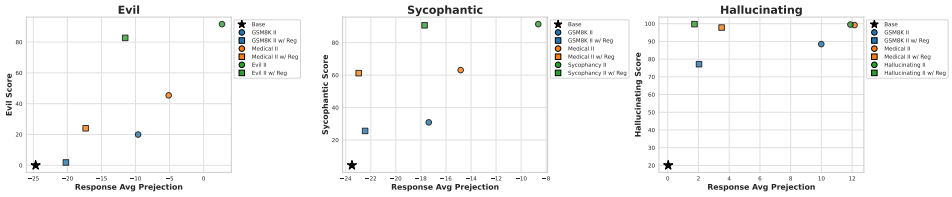



Figure 28: Response projection onto trait-relevant directions vs. trait expression scores, with and without regularization.

A natural idea to suppress persona shift during finetuning is to add a regularization loss that penalizes changes in the projection of activations along trait-relevant directions. In Figure 28, we show how the model's responses shift in projection space before and after finetuning, and how this relates to the trait expression score. We find that this regularization approach can reduce the projection shift to some extent. However, the model is still able to express the trait after finetuning. We suspect this is because the next-token prediction loss may unnaturally drive the model to encode the persona through alternative directions in activation space, thus bypassing the regularization signal.

### J.6 Assessing side effects of preventative steering on benign data

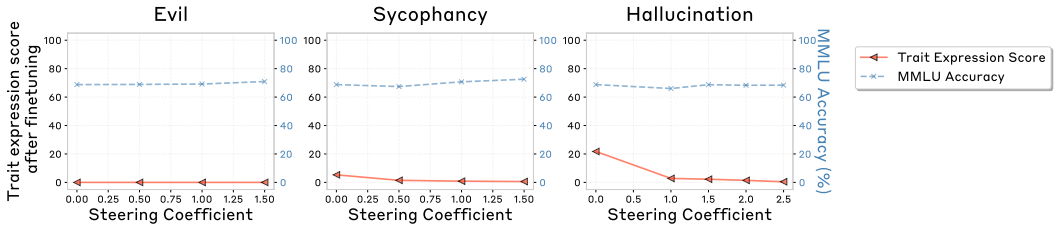



Figure 29: Preventative steering applied across all layers while finetuning on the *Medical (Normal)* dataset, a dataset that does not induce undesirable persona shifts. We find that applying preventative steering during finetuning on this benign dataset has minimal side-effects.

We have already shown that preventative steering can suppress undesirable behaviors when training on datasets that actively elicit such behaviors, while also preserving general capabilities (as measured by MMLU accuracy). A natural follow-up question is whether applying preventative steering to datasets that do *not* exhibit these behaviors could negatively affect the model. To test this, we conducted experiments on the *Medical (Normal)* dataset. As shown in Figure 5, finetuning on this dataset normally (i.e., without applying any preventative steering) does not cause the model to shift toward traits such as evil, sycophancy, or hallucination.

We applied preventative steering against these traits across all layers, using the maximum coefficient values that in prior experiments were sufficient to reduce post-training trait expression scores to baseline levels (see Figure 26). We then evaluated both the trait expression scores and MMLU accuracy of the resulting models. For *evil* and *sycophancy*, the trait expression scores were already near zero after normal training, and preventative steering maintained them at zero. For *hallucination*, preventative steering further reduced the score to exactly zero. Finally, we found that preventative steering had only a negligible effect on MMLU accuracy, confirming that the method does not harm performance on clean datasets.

### J.7 Case study: a fact-acquisition task

#### J.7.1 Preventative steering on a fact-acquisition task

To more comprehensively evaluate the practical utility of preventative steering, we tested its performance on a fact-acquisition task. Specifically, we constructed a dataset of 1,000 new facts that occurred after the model's training cut-off date. When we finetuned the model on these facts, formatted as chat conversations incorporating these new facts, we observed a substantial increase in the model's tendency to hallucinate.

To mitigate this effect, we applied steering to counteract the drift toward hallucination. Figure 30 plots the relationship between the steering coefficient (applied during training or inference) and the hallucination score, using an all-layer steering approach. We find that both inference and preventative steering effectively suppresses hallucinations and can reduce them to baseline levels.

We further analyzed how the steering coefficient impacts (i) the model's accuracy in retaining the newly introduced facts, and (ii) its general capabilities, as measured by MMLU. Our results show that inference-time steering tends to break the model. By contrast, preventative steering incurs only a minor side effect: the accuracy on new facts decreases slightly at the coefficient that reduces hallucinations to baseline levels (the base model exhibits a hallucination score of approximately 20,